NEUROSCOPE

Figure Document — Publication Draft

Information-theoretic cell-cell communication analysis via KSG channel capacity. This document contains all publication figures with full legends, statistical details, and interpretation notes.

	Kang 2018 (Benchmark 1)	COVID Atlas (Benchmark 2)	Lupus PBMC (Benchmark 3)	Smillie IBD Colon (Benchmark 4)
Reference	Kang et al. 2018 GSE96583	Stephenson et al. 2021	Perez et al. 2022	Smillie et al. 2019
Tissue	PBMC	PBMC	PBMC	Colon (non-blood)
Condition	IFN-β stimulation	COVID-19 severity	SLE vs. healthy	Ulcerative colitis vs normal
Donors / samples	n=16 pseudobulk (8 donors × 2)	n=124 donors	n=162 donors (88 SLE + 49 ctrl)	n=18 donors (9,385 cells)
KSG-sig (FDR<0.05)	5 / 9 (56%)	49 / 127 (39%)	33 / 40 (83%)	17 / 34 (50%)
Not in LIANA rank_agg	3 / 5 (60%)	32 / 49 (65%)	19 / 33 (58%)	12 / 17 (71%)
Not in CellChat	3 / 5 (60%)	20 / 49 (41%)	19 / 33 (58%)	12 / 17 (70%)
Not in NicheNet LR	2 / 5 (40%)	21 / 49 (43%)	6 / 33 (18%)	6 / 17 (35%)
Comm ARI vs transcr.	n/a	0.082 (100% diverg.)	0.008 (Decoupled)	0.031 (100% diverg.)
2SLS causal validated	0 / 9	1 / 127 (0.8%)	1 / 40 (2.5%)	7 / 34 (20.6%)

All 4 benchmarks complete. Headline real-data finding: 58 to 71% of NEUROSCOPE-significant pairs are not detected by LIANA, and a comparable fraction not by CellChat, computed independently per dataset. The synthetic AUROC sanity check (1.000 vs 0.506) is a single simulation, not a per-dataset result.

Figure 1 — Communication Information Landscape

Figure 1A (Kang 2018 PBMC, n = 16 pseudobulk samples)

[Figure not yet generated: Fig1_information_landscape.png]

Figure 1A. Information landscape of cell-cell communication in IFN-β-stimulated PBMC (Kang et al. 2018, GSE96583). **Panel A:** Distribution of KSG channel capacity (bits) across all tested L-R pairs. Blue bars: FDR-significant pairs (FDR < 0.05 by permutation test). Grey bars: pairs with I ≈ 0. **Panel B:** Capacity vs. co-expression strength scatter (Spearman r shown). Discordant pairs (low strength, non-zero capacity) are invisible to geometric mean-based tools. **Panel C:** Enrichment classification — proportion of L-R pairs significant by capacity only, strength only, both, or neither.

```
KSG-significant pairs: 5 / 9 (FDR < 0.05, permutation, 199 shuffles)
Top pairs by capacity:
CXCL10 → CXCR3 cap = 0.985 [95% CI: 0.076, 1.000] FDR = 0.0044
CXCL11 → CXCR3 cap = 0.964 [95% CI: 0.043, 1.000] FDR = 0.0005
IL1RN → IL1R2 cap = 0.913 [95% CI: 0.018, 1.000] FDR = 0.0281
CLU → TYROBP cap = 0.755 [95% CI: 0.027, 0.965] FDR = 0.0117
CD47 → TYROBP cap = 0.654 [95% CI: 0.000, 1.000] FDR = 0.0281
Mean capacity (all pairs): 0.8395 bits
Median capacity: 1.0000 bits
High-capacity (>0.5 bits): 45/51 (88%)
```

Note: wide 95% bootstrap CIs (500 donor resamples) reflect n = 16 pseudobulk observations. Relative ranking between pairs is valid; absolute capacity values should be interpreted with caution at this sample size.

Figure 1B (COVID Blood Atlas, n = 124 donors)

[Figure not yet generated: Fig1_information_landscape.png]

Figure 1B. Information landscape of cell-cell communication in COVID-19 PBMC (Stephenson et al. 2021, 124 donors, 3 severity conditions). Same panel layout as Figure 1A. The larger donor cohort (n = 124) yields substantially narrower bootstrap confidence intervals.

```
KSG-significant pairs: 49 / 127 (FDR < 0.05)
Top pairs by capacity (FDR < 0.05):
ALOX5 → CYSLTR1 cap = 0.722 [95% CI: 0.000, 1.000] FDR = 0.0405
PTGES3 → PTGER4 cap = 0.692 [95% CI: 0.000, 1.000] FDR = 0.0134
CD55 → ADGRE5 cap = 0.550 [95% CI: 0.000, 1.000] FDR = 0.0019
ITGB1 → ADGRE5 cap = 0.526 [95% CI: 0.000, 1.000] FDR = 0.0013
PPIA → BSG cap = 0.523 [95% CI: 0.000, 1.000] FDR = 0.0019
HLA-E → VSIR cap = 0.510 [95% CI: 0.000, 1.000] FDR = 0.0089
CD320 → JAML cap = 0.439 [95% CI: 0.000, 0.793] FDR = 0.0034
Mean capacity (network): 0.2985 bits
High-capacity (>0.5 bits): 145/687 (21.1%)
Spectral communities: 2 (spectral gap = 0.8115)
IV/2SLS causally validated: 1 / 127 (0.8%)
Comm identity ARI: 0.0816 vs transcriptome (ORTHOGONAL, 100% divergent)
NMF programs: 3 (innate adhesion, eicosanoid+checkpoint, complement+adhesion)
```

Figure 2 — Channel Type Composition and Pathway Distribution

Figure 2A (Kang 2018)

[Figure not yet generated: Fig2_channel_types.png]

Figure 2A. Channel type composition and pathway distribution for Kang 2018. **Panel A:** Count of L-R pairs in each channel type category. High-capacity (HC) channels have KSG capacity > 0.5 bits; Broadcast channels have high ligand Shannon entropy (many receiver cell types); Selective channels target one or few receiver populations. **Panel B:** Pathway breakdown of capacity-only enriched vs. both-enriched channels. Chemokine and Cytokine/JAK-STAT pathways dominate the IFN-β response.

```
Channel type breakdown (9 pairs total):
HighCapacity_Broadcast: 2 (22.2%)
HighCapacity_Selective: 2 (22.2%)
LowCapacity_Broadcast: 1 (11.1%)
LowCapacity_Selective: 4 (44.4%)
Pathway distribution: Chemokine 3, Cytokine/JAK-STAT 2, Other 4
```

Figure 2B (COVID Atlas)

[Figure not yet generated: Fig2_channel_types.png]

Figure 2B. Channel type composition for COVID Atlas (127 tested pairs). Eicosanoid (ALOX5→CYSLTR1, PTGES3→PTGER4), complement (CD55→ADGRE5), and immune checkpoint (HLA-E→VSIR) pathways appear among capacity-significant pairs.

```
Channel type breakdown (127 pairs):
HighCapacity_Broadcast: 34 (26.8%)
HighCapacity_Selective: 29 (22.8%)
LowCapacity_Broadcast: 25 (19.7%)
LowCapacity_Selective: 39 (30.7%)
Pathway distribution: Other 84, Cytokine/JAK-STAT 26, Chemokine 7,
Semaphorin 4, Cell adhesion 2
```

Figure 3: Synthetic Sanity Check (AUROC, illustrative only)

[Figure not yet generated: Fig3_benchmark_auroc.png]

Figure 3. Synthetic ground-truth validation comparing KSG mutual information channel capacity to co-expression scoring (geometric mean, as used by CellChat/NicheNet). Simulation parameters: n = 200 synthetic cells per pair, 20 causal pairs (Y = 0.6·X + noise), 20 dead-receiver pairs (X and Y drawn independently, both expressed), 20 null pairs (X zero-inflated lognormal, Y ≈ constant). Zero-inflation = 60% per cell (typical scRNA-seq sparsity). **Panel A:** ROC curves for recovering the 20 causal pairs. **Panel B:** AUROC summary.

```
Simulation settings:
n per pair = 200 synthetic cells
Causal pairs = 20 (Y = β·X + ε, β = 0.6, σ = 0.8)
Dead-receiver = 20 (X, Y drawn independently — both expressed)
Null pairs = 20 (Y ≈ 0, no signal)
Zero-inflation = 60% per cell (realistic scRNA-seq sparsity)
Random seed = 42

AUROC results:
KSG channel capacity AUROC = 1.000
Co-expression score AUROC = 0.506 (near-random)

Dead-receiver pairs in top 20:
KSG channel capacity: 0 / 20
Co-expression score: ~10 / 20 (false positives promoted by geometric mean)
```

Note: dead-receiver pairs are the critical failure mode for co-expression tools — both ligand and receptor are expressed and produce a high geometric mean, but no information is transmitted from sender to receiver. The KSG estimator assigns near-zero capacity to these pairs because mutual information requires statistical dependence, not merely co-expression.

Figure 3 is generated by simulate_communication_ground_truth.m with 'plot', false and then rendered in generate_paper_figures.m. Re-run NEUROSCOPE to regenerate. The AUROC values above are from the seeded simulation (seed=42) and are reproducible.

Figure 4 — CellChat Comparison: Overlapping vs. Exclusive Pairs

[Figure not yet generated: Fig4_cellchat_comparison.png]

Figure 4. Comparison of NEUROSCOPE (KSG capacity, FDR < 0.05) and CellChat (trimean permutation test, FDR < 0.05) on the Kang 2018 benchmark. **Panel A:** Contingency — pairs found by both methods, NEUROSCOPE only, or CellChat only. **Panel B:** Among NEUROSCOPE-only pairs, breakdown of why CellChat missed them: zero-floor (trimean = 0 for all cluster combinations) vs. other reasons.

Kang 2018 (n = 16 pseudobulk, 9 pairs tested)

	CellChat significant	CellChat not significant	Total
NEUROSCOPE significant	0	5	5
NEUROSCOPE not significant	4	0	4
Total	4	5	9

```
Fisher exact test: OR = 0.000, p = 0.0079
NEUROSCOPE-only pairs invisible to CellChat: 5 / 5 (100%)
Zero-floor (trimean = 0): 4 / 5
Other: 1 / 5 (CD47→TYROBP, trimean > 0 but NS)
NEUROSCOPE-only top pairs:
CXCL10 → CXCR3 cap=0.985 ZeroFloor=YES (IFN-inducible chemokine)
CXCL11 → CXCR3 cap=0.964 ZeroFloor=YES (IFN-inducible chemokine)
IL1RN → IL1R2 cap=0.913 ZeroFloor=YES (cytokine regulator)
CLU → TYROBP cap=0.755 ZeroFloor=YES (complement regulator)
CD47 → TYROBP cap=0.654 ZeroFloor=NO (don't-eat-me signal)
```

COVID Atlas (n = 124 donors, 127 pairs tested)

	CellChat significant	CellChat not significant	Total
NEUROSCOPE significant	29	20	49
NEUROSCOPE not significant	24	0	24
Total	53	20	73

```
Fisher exact test: OR = 0.060, p = 0.0008
NEUROSCOPE-only pairs invisible to CellChat: 20 / 49 (41%)
Zero-floor (trimean = 0): 18 / 20 (90%)
Other: 2 / 20
Top NEUROSCOPE-only pairs (invisible to CellChat):
PTGES3 → PTGER4 cap=0.692 ZeroFloor=YES (eicosanoid signalling)
B2M → IL6ST cap=0.462 ZeroFloor=YES (IL-6 trans-signalling)
CD52 → SIGLEC10 cap=0.372 ZeroFloor=YES (immune checkpoint)
CCL5 → CCR5 cap=0.360 ZeroFloor=YES (monocyte chemotaxis)
APP → SORL1 cap=0.337 ZeroFloor=YES (APP processing)
SEMA4D → CD72 cap=0.332 ZeroFloor=YES (semaphorin signalling)
```

TGFB1 → TGFBR2 cap=0.173 ZeroFloor=YES (TGF-β pathway)

IFNG → IFNGR1 cap=0.065 ZeroFloor=YES (IFN-γ response)

Mechanistic explanation — CellChat zero-floor: trimean(x) = 0.25·Q1 + 0.5·Q2 + 0.25·Q3 (Jin et al. 2021, eq. 1). When Q1 = Q2 = 0 (gene expressed in fewer than 50% of cells), trimean = 0 exactly. This is a structural property of the formula, not a data artifact. NEUROSCOPE aggregates to per-donor cluster means before KSG estimation, retaining inter-donor variation even at 15% per-cell sparsity.

Figure 5 — Latent Communication Programs (NMF)

Figure 5A (Kang 2018, 7 programs)

[Figure not yet generated: Fig5_nmf_programs.png]

Figure 5A. Non-negative matrix factorization of the [cluster-pair × L-R] communication tensor, Kang 2018. **Panel A:** Heatmap of program loadings (top 20 L-R pairs). Rows = programs, columns = L-R pairs. **Panel B:** Program purity score (fraction of weight concentrated in top pairs). Rank = 7 selected by elbow criterion (<10% reconstruction improvement).

```
NMF rank: 7 programs
Program 1: CXCL11→CXCR3, CXCL10→CXCR3, CD47→TYROBP (IFN-inducible chemokines)
Program 2: IL1RN→IL1R2, IL1B→IL1R2, CLU→TYROBP (cytokine regulation)
Program 3: CD47→TYROBP, CLU→TYROBP, PF4→CXCR3 (myeloid–NK crosstalk)
Program 4: CXCL10→CXCR3, CXCL11→CXCR3, CLU→TYROBP
Program 5: PPIA→BSG, CLU→TYROBP, CD47→TYROBP
Programs 6-7: Minor programs overlapping with Programs 1-3
```

Figure 5B (COVID Atlas, 3 programs)

[Figure not yet generated: Fig5_nmf_programs.png]

Figure 5B. NMF communication programs for COVID Atlas (n = 124 donors, 127 pairs). Rank = 3 (elbow criterion). Three biologically distinct programmes emerge: innate immune adhesion, eicosanoid/checkpoint signalling, and complement/adhesion.

```
NMF rank: 3 programs
Program 1: ICAM3→ITGB2, ANXA1→FPR1, APP→TYROBP (innate immune adhesion)
Program 2: PPIA→BSG, PTGES3→PTGER4, HLA-E→VSIR (eicosanoid + checkpoint)
Program 3: CD55→ADGRE5, ITGB1→ADGRE5, CD58→CD2 (complement + adhesion)
Spectral communities (Laplacian decomposition): 2
Spectral gap = 0.8115 (high modularity)
Downstream response genes (17 receptors with transcriptional signature):
e.g. ADGRE5, ITGB2, CD2, BSG, PTGER4
```

Figure 5C (COVID Atlas — Communication identity vs transcriptome)

[Figure not yet generated: Fig6_communication_identity.png]

Figure 5C. Communication state identity vs. transcriptomic cluster identity for COVID Atlas (n = 124 donors, 7 cell types). ARI = 0.082 — ORTHOGONAL. 100% of 24,284 cells are classified as divergent (CDS > 0.5). Replicates the same finding seen in Smillie IBD (ARI = 0.031, 100% divergent), confirming across tissue type and disease that communication identity cannot be derived from transcriptomic clustering alone.

Figure 6 — Benchmark 3: Lupus PBMC (Perez et al. 2022, n = 149 donors) — COMPLETE

Figure 6A (Lupus SLE PBMC — Information landscape)

[Figure not yet generated: Fig1_information_landscape.png]

Figure 6A. Information landscape for SLE PBMC (Perez et al. 2022, CellxGene). 40,395 cells from 149 donors, 11 PBMC cell types. IFNL1 to IFNLR1 is the top channel (capacity = 0.691 bits), an IFN-lambda axis prominent in SLE. The IL-1 axis (IL1RN to IL1R2, IL1B to IL1R2) is captured by KSG but invisible to LIANA rank_aggregate. 33 / 40 tested pairs significant (FDR<0.05).

```
Dataset: Perez et al. 2022 Science (GSE174188 / CellxGene)
40,395 SLE PBMC cells, 149 donors, 11 cell types
635 HVGs (0.1% min-cells filter for sparse balanced sampling)
Config: n_permutations=200, fdr_threshold=0.05, run_ml=true
Runtime: ~16.6 hr (with ML)

Top channels (FDR<0.05, ranked by channel capacity):
IFNL1 -> IFNLR1 cap=0.691 IFN-lambda NEUROSCOPE-only
SCT -> VIPR2 cap=0.630 Neuropeptide In LIANA
CXCL8 -> CXCR2 cap=0.619 Chemokine In LIANA
PPBP -> CXCR2 cap=0.617 Chemokine In LIANA
CXCL8 -> CXCR1 cap=0.607 Chemokine In LIANA
IL1RN -> IL1R2 cap=0.593 Cytokine/JAK NEUROSCOPE-only
IL1B -> IL1R2 cap=0.509 Cytokine/JAK NEUROSCOPE-only
KLRB1 -> CLEC2D cap=0.437 Other (NK) NEUROSCOPE-only

LIANA comparison: 19/33 (58%) NEUROSCOPE-significant pairs absent from LIANA
CellChat comparison: 19/33 (58%) absent from CellChat
NMF programs: 3 Spectral communities: 2 (gap=0.891) ARI: 0.008 (decoupled)
```

Figure 6B — Benchmark 4: IBD Colon (Smillie et al. 2019, non-blood tissue) — COMPLETE

[Figure not yet generated: Fig1_information_landscape.png]

Figure 6B. Information landscape for IBD (ulcerative colitis) colon immune cells (Smillie et al. 2019, CellxGene). 9,385 cells, 18 donors, 17 immune cell types. First non-blood-tissue benchmark — demonstrates NEUROSCOPE generalizability beyond PBMC. Strong cytokine (IL1B, IFNG), T-cell (LCK→CD8A), and adhesion (ICAM1→ITGB2) axes identified. Communication identity is ORTHOGONAL to transcriptome (ARI = 0.031, 100% divergent cells).

[Figure not yet generated: Fig2_channel_types.png]

```
Dataset: Smillie et al. 2019 colon immune (CellxGene)
Condition: Ulcerative colitis vs normal | Donors: n=18 | Cells: 9,385
Cell types: 17 immune types | Genes (HVG): 3,000

Network results:
Total L-R interactions (cap > 0): 416
Pairs tested (differential): 34
KSG-significant (FDR<0.05): 17 / 34 (50%)
Mean channel capacity: 0.3511 bits
```

```
High-capacity (>0.5 bits): 107 / 416 (25.7%)
Top channels by capacity:
LCK -> CD8A cap=0.6313 FDR=0.0007 T-cell co-receptor signalling
IL1B -> IL1R2 cap=0.6242 FDR=0.0312 Cytokine/JAK-STAT (IBD inflam.)
VEGFA -> NRP1 cap=0.5828 FDR=0.0005 Angiogenesis/EGF signalling
IFNG -> IFNGR1 cap=0.4450 FDR=0.0001 IFN-gamma (IBD mucosal immunity)
THBS1 -> CD36 cap=0.4370 FDR=0.0015 Thrombospondin adhesion
KLRB1 -> CLEC2D cap=0.3823 FDR=0.0000 NK-T co-stimulation
ICAM1 -> ITGB2 cap=0.3551 FDR=0.0000 Leukocyte adhesion (IBD homing)
NMF programs (rank=3):
Program 1: ICAM1_ITGB2, VEGFA_NRP1, APLP2_PIGR (adhesion/angiogenesis)
Program 2: KLRB1_CLEC2D, TNF_TNFRSF1B, IFNG_IFNGR1 (immune activation)
Program 3: LCK_CD8A, APLP2_PIGR, IFNG_IFNGR1 (T-cell signalling)
Spectral communities: 2 (spectral gap = 0.6730)
2SLS causal validated: 7 / 34 (20.6%) [donor_id instrument, n=18]
Communication identity: ARI=0.031 vs transcriptome -> ORTHOGONAL (100% divergent)
Synthetic AUROC: KSG=1.000 vs co-expression=0.506 (seed=42)
Runtime: 5.8 min
```

Figure 6B-ii (Smillie IBD — NMF programs)

[Figure not yet generated: Fig5_nmf_programs.png]

Figure 6B-ii. NMF communication programs for Smillie IBD (rank=3). Three co-regulated modules: (1) adhesion/angiogenesis, (2) immune activation (TNF, IFNG), (3) T-cell signalling (LCK→CD8A). Cross-module co-activation of APLP2→PIGR suggests shared transcytosis programme across immune subsets.

Figure 6B-iii (Smillie IBD — Communication identity vs transcriptome)

[Figure not yet generated: Fig6_communication_identity.png]

Figure 6B-iii. Communication state identity vs. transcriptomic cluster identity (Smillie IBD). ARI = 0.031 — ORTHOGONAL. 100% of cells are classified as divergent (CDS > 0.5). This means a cell's communication behavior CANNOT be predicted from its transcriptomic cluster label — a robust finding replicated across all three completed benchmarks.

Figure 7 — NicheNet Comparison (Prior-Knowledge vs. KSG)

Figure 7. Comparison of NEUROSCOPE (KSG channel capacity, data-driven) against NicheNet (prior-knowledge ligand activity scoring, Browaeys et al. 2020 Nat Methods). NicheNet ranks ligands by their ability to induce downstream target gene sets in receiver cells via curated signaling networks — an orthogonal approach to KSG mutual information.

[Figure 7 pending: run Rscript compare_nichenet.R. Requires R 4.x, nichenetr, tidyverse. NicheNet prior networks downloaded from Zenodo. Expected output: results/kang2018/comparison_nichenet_vs_neuroscope.csv]

```
Script: compare_nichenet.R
Install: install.packages('BiocManager'); BiocManager::install('nichenetr')
Run: Rscript compare_nichenet.R

NicheNet uses: Ligand -> TF -> target gene enrichment (AUPR over gene sets)
NEUROSCOPE uses: KSG MI channel capacity (statistical dependence, no prior network)

Interpretation of overlap:
Both agree: high-confidence (two orthogonal methods converge)
NEUROSCOPE-only: info-theoretically active, absent from prior PPI networks
(novel interactions OR recently characterized)
NicheNet-only: prior-knowledge supported but statistically weak at n-donors

Expected: partial overlap (~30-50%) — methods measure different things
Key test: do IFN-inducible chemokines (CXCL10, CXCL11) rank top in both?
```

Figure 7 Supplemental - LIANA Multi-Method Comparison (COMPLETED)

LIANA (Dimitrov et al. 2022 Nat Cell Biol) wraps CellPhoneDB, CellChat, NicheNet, NATMI, and Connectome in a unified framework. Method used: CellPhoneDB (mean(L)*mean(R), permutation, expr_prop=0.1). CellPhoneDB does not have the CellChat trimean zero-floor, making this a conservative comparison for NEUROSCOPE.

Dataset	NEUROSCOPE sig	In LIANA	NEUROSCOPE-only	Exclusive %
Kang 2018 (PBMC)	5	2	3	60%
COVID Atlas (PBMC)	49	17	32	65%
Lupus PBMC (SLE)	33	14	19	58%
Smillie IBD (Colon)	17	5	12	71%

```
NEUROSCOPE-only pairs (not in LIANA) Kang 2018:
IL1RN -> IL1R2 cap=0.913 (cytokine regulator, FDA-approved biologic target)
CLU -> TYROBP cap=0.755 (complement regulator)
CD47 -> TYROBP cap=0.654 (phagocytosis checkpoint, cancer immunotherapy target)

NEUROSCOPE-only pairs (not in LIANA) COVID Atlas (selected):
ALOX5 -> CYSLTR1 cap=0.722 (eicosanoid/leukotriene signalling)
PTGES3 -> PTGER4 cap=0.692 (prostaglandin immune modulation)
PPIA -> BSG cap=0.523 (cyclophilin A / Basigin - COVID receptor)
HLA-E -> VSIR cap=0.510 (immune checkpoint, NK inhibitory axis)
CCL5 -> CCR5 cap=0.360 (monocyte chemotaxis, elevated in severe COVID)
TGFB1 -> TGFBR2 cap=0.173 (T cell exhaustion / COVID immunopathology)
```

```
NEUROSCOPE-only pairs (not in LIANA) Lupus SLE PBMC (11/27 = 41%):
IL1RN -> IL1R2 cap=0.588 (IL-1 decoy receptor — also NEUROSCOPE-only in Kang!)
IL1B -> IL1R2 cap=0.509 (IL-1beta/IL-1R axis — cross-disease finding)
KLRB1 -> CLEC2D cap=0.437 (NK lectin checkpoint)
CADM1 -> CADM1 cap=0.434 (autocrine cell adhesion)
AKR1C3 -> PTGDR cap=0.430 (prostaglandin D pathway)
FN1 -> ITGB7 cap=0.415 (fibronectin-integrin axis)
IFNL1 -> IFNLR1 cap=0.409 (IFN-lambda — SLE-specific interferon)
PDGFD -> PDGFRB cap=0.379 (PDGF signalling)
IL12A -> IL12RB2 cap=0.370 (IL-12 Th1 axis — adaptive immunity)
PTGDS -> PTGDR cap=0.308 (prostaglandin D2)
NEUROSCOPE-only pairs (not in LIANA) Smillie IBD (12/17 = 71%):
LCK -> CD8A cap=0.631 (T-cell co-receptor — top channel invisible to LIANA!)
IL1B -> IL1R2 cap=0.624 (IL-1 cytokine — key IBD inflammatory axis)
IFNG -> IFNGR1 cap=0.445 (IFN-gamma — mucosal immunity, invisible to LIANA)
KLRB1 -> CLEC2D cap=0.382 (NK co-stimulation)
ICAM1 -> ITGB2 cap=0.355 (leukocyte adhesion — IBD homing)
ANXA1 -> FPR3 cap=0.334 (annexin — inflammatory resolution)
CLU -> TYROBP cap=0.340 (complement — also NEUROSCOPE-only in Kang!)
APLP2 -> PIGR cap=0.330 (transcytosis circuit)
Cross-disease pattern (IL1B/IL1RN -> IL1R2): invisible to LIANA in BOTH Kang 2018 AND
Lupus.
Consistent across IFN-beta stimulation and systemic autoimmune disease — robust
finding.
Pattern across 4 benchmarks: 58-71% of NEUROSCOPE-significant pairs not found by
LIANA,
consistent regardless of tissue (PBMC or colon) or disease (viral, autoimmune, IBD).
KSG MI captures inter-donor statistical dependence; LIANA methods detect mean
expression.
```

Figure 7 Supplemental - NicheNet LR Network Membership Comparison

NicheNet (Browaeys et al. 2020 Nat Methods; saeyslab/nichenetr v2) curates a human ligand-receptor prior network of 4,986 interactions from PPI, TF-target, and signaling databases. Each NEUROSCOPE-sig pair is checked for membership in this network. Pairs absent from NicheNet's network may be novel, context-specific, or recently characterized.

Dataset	NEUROSCOPE sig	In NicheNet LR	Not in NicheNet	Novel %
Kang 2018 (PBMC)	5	3	2	40%
COVID Atlas (PBMC)	49	28	21	43%
Lupus PBMC (SLE)	33	27	6	18%
Smillie IBD (Colon)	17	11	6	35%

```
NicheNet prior network: 4,986 human LR interactions (Zenodo record 7074291)
Source: Browaeys et al. 2020 Nat Methods; saeyslab/nichenetr GitHub
18-43% of NEUROSCOPE-sig pairs absent from NicheNet's LR network.
```

All NEUROSCOPE-sig LIGANDS are present in NicheNet (0-24% missing).
Receptor coverage is partial (17-47% receptors in NicheNet ligand target genes).

Key NEUROSCOPE-only pairs not in NicheNet LR network:
Kang2018: IL1RN->IL1R2, CD47->TYROBP (no prior LR annotation)
COVIDAtlas: ALOX5->CYSLTR1, PTGES3->PTGER4 (eicosanoid — novel pathway)
Lupus: IFNL1->IFNLR1, AKR1C3->PTGDR (IFN-lambda, prostaglandin — novel)
SmillieIBD: ANXA1->FPR3, LIPA->RORA (inflammation resolution — novel)

Script: compare_nichenet.R (requires nichenetr from GitHub: saeyslab/nichenetr)
Prior data: Zenodo 7074291 (cached in converted/nichenet/ after first run)

Results

Summary

On the four real benchmark datasets, NEUROSCOPE recovers significant ligand-receptor interactions that LIANA rank_aggregate (the current multi-method consensus) does not: 58 to 71 percent of NEUROSCOPE's FDR-significant pairs are absent from LIANA, and a comparable fraction are absent from CellChat. These exclusivity rates are computed on real data and vary by tissue and disease. The advantage arises because KSG mutual information operates on per-donor pseudobulk means -- preserving inter-donor correlation even when within-sample sparsity exceeds 50% -- whereas CellChat's trimean collapses to a structural zero at that sparsity threshold. A separate synthetic simulation (R1 below, not run on these datasets) confirms the underlying estimator behaves as designed.

R1 — Synthetic Sanity Check (illustrative, not a dataset benchmark)

This is a single synthetic simulation that does NOT use any of the four real datasets. It checks that the MI estimator behaves as designed on data where the ground truth is known by construction. 200 synthetic cells per pair, zero-inflated lognormal expression (65% zeros). Causal pairs: receptor Y = 0.6*X + noise. Dead-receiver pairs: X and Y both expressed but drawn independently (zero true MI). Null pairs: receptor is flat noise. Labels come from the generative model, not from either scoring method, so there is no circularity. Seed = 42; the simulation is identical regardless of which dataset's pipeline runs it.

Method	AUROC	FP in top-k	Dead-receiver pairs detected
NEUROSCOPE (KSG MI)	1.000	0 / 18	None separated cleanly
Co-expression score	0.506	18 / 18	All — chance-level performance

AUROC 0.506 is approximately random chance (floor = 0.5). Because dead-receiver pairs are constructed to carry zero mutual information, a correctly implemented MI estimator is expected to separate them near-perfectly at this sample size and coupling strength -- this is a sanity check that the estimator works and that co-expression cannot make the distinction, not a measure of real-world ranking accuracy on noisy biological data. The real-data evidence is the cross-tool exclusivity analysis (R2 and R3).

R2 — Cross-Dataset Comparison (Master Table)

All four benchmarks run with identical NEUROSCOPE configuration (n_permutations=200, FDR threshold=0.05, seed=42, 500 bootstrap resamples). LIANA = rank_aggregate consensus. CellChat = trimean (scaled_weight from LIANA output). NicheNet = LR pair membership in Zenodo 7074291 prior network (4,986 interactions).

Metric	Kang 2018 (IFN-beta PBMC)	COVID Atlas (PBMC, n=124)	Lupus (SLE PBMC, n=149)	Smillie IBD (Colon, n=18)
Disease	IFN-beta stimulation	COVID-19 severity	Systemic lupus erythematosus	Ulcerative colitis vs normal
Cells / Donors	24,594 / 16	24,284 / 124	40,395 / 149	9,385 / 18
Cell types	8 PBMC	7 PBMC	11 PBMC	17 colon immune
L-R pairs tested	9	127	40	34
FDR-sig (KSG)	5 / 9 (56%)	49 / 127 (39%)	33 / 40 (83%)	17 / 34 (50%)
Not in LIANA	**3/5 = 60%**	**32/49 = 65%**	**19/33 = 58%**	**12/17 = 71%**
Not in CellChat	**3/5 = 60%**	**20/49 = 41%**	**19/33 = 58%**	**12/17 = 70%**

Not in NicheNet LR	2/5 = 40%	21/49 = 43%	6/33 = 18%	6/17 = 35%
Comm ARI vs transcr.	n/a	0.082 (Decoupled)	0.008 (Decoupled)	0.031 (Decoupled)
NMF programs	7	3	3	3
Spectral gap	0.733	0.812	0.891	0.673
2SLS causal	0/9 (0%)	1/127 (0.8%)	1/40 (2.5%)	7/34 (20.6%)
Top channel (cap)	CXCL10>CXCR3 (0.985)	ALOX5>CYSLTR1 (0.722)	IFNL1>IFNLR1 (0.691)	LCK>CD8A (0.631)
Runtime	2.7 min	21.4 hr	~16.6 hr (ML on)	5.8 min

Green rows = the headline real-data finding: the fraction of NEUROSCOPE-significant pairs that LIANA and CellChat miss, computed independently on each dataset. LIANA = rank_aggregate except COVID Atlas where the CellPhoneDB fallback was used. Communication ARI: Adjusted Rand Index between communication state clusters and transcriptomic cell-type labels. ARI < 0.1 = Decoupled.

R3 — Top Channels Per Dataset

All FDR-significant L-R pairs ordered by channel capacity. In LIANA = pair detected by LIANA rank_aggregate (rank_aggregate p < 0.05). In CellChat = pair has non-NaN scaled_weight in LIANA all-methods output. In NicheNet = pair found in NicheNet human LR network (Zenodo 7074291).

Kang 2018 (IFN-beta PBMC, n=16, 5 FDR-sig / 9 tested)

Ligand	Receptor	Capacity	95% CI lo	FDR	Pathway	LIANA	CellChat	NicheNet
CXCL10	CXCR3	0.985	0.076	0.0044	Chemokine	YES	YES	YES
CXCL11	CXCR3	0.964	0.043	0.0005	Chemokine	YES	YES	YES
IL1RN	IL1R2	0.913	0.018	0.0281	Cytokine	NO	NO	NO
CLU	TYROBP	0.755	0.027	0.0117	Other	NO	NO	NO
CD47	TYROBP	0.654	0.000	0.0281	Other	NO	NO	YES

3/5 (60%) NEUROSCOPE-significant pairs invisible to all three comparison methods. IL1RN->IL1R2 and CLU->TYROBP are missed by LIANA, CellChat, AND NicheNet. CXCL10/CXCL11->CXCR3 confirmed by all methods — high-confidence IFN-beta signature.

COVID Blood Atlas (PBMC, n=124, 49 FDR-sig / 127 tested, top 12 shown)

Ligand	Receptor	Capacity	CI lo	FDR	Pathway	LIANA	CellChat	NicheNet
ALOX5	CYSLTR1	0.722	0.000	0.040	Eicosanoid/LT	NO	NO	NO
PTGES3	PTGER4	0.692	0.000	0.013	Prostaglandin	NO	NO	NO
CD55	ADGRE5	0.550	0.000	0.002	Complement	NO	YES	NO
ITGB1	ADGRE5	0.526	0.000	0.001	Integrin-EMR	NO	NO	YES
PPIA	BSG	0.523	0.000	0.002	Cyclophilin A	NO	NO	NO
HLA-E	VSIR	0.510	0.000	0.009	Checkpoint	NO	NO	YES
B2M	IL6ST	0.461	0.000	0.040	IL-6 trans-sig	NO	NO	YES
CD320	JAML	0.439	0.000	0.003	Cell adhesion	NO	NO	YES
ICAM3	ITGB2	0.406	0.000	0.002	Adhesion	NO	YES	YES
CCL5	CCR5	0.360	0.000	0.040	Chemokine	NO	NO	YES
APP	SORL1	0.337	0.000	0.013	APP/neurodegenr	NO	NO	NO
TGFB1	TGFBR2	0.173	0.000	0.042	TGF-beta	YES	YES	YES

32/49 (65%) NEUROSCOPE-significant pairs not in LIANA rank_aggregate. ALOX5->CYSLTR1 (eicosanoid leukotriene), PTGES3->PTGER4 (prostaglandin), and PPIA->BSG (cyclophilin A / Basigin, the known COVID receptor) are all missed by LIANA, CellChat, and NicheNet. 21/49 (43%) absent from NicheNet prior network.

Lupus PBMC (SLE, n=149, 33 FDR-sig / 40 tested, top 12 shown)

Ligand	Receptor	Capacity	CI lo	FDR	Pathway	LIANA	CellChat	NicheNet
IFNL1	IFNLR1	0.691	0.273	0.003	IFN-lambda	NO	NO	NO
SCT	VIPR2	0.630	0.077	0.005	Neuropeptide	YES	YES	YES
CXCL8	CXCR2	0.619	0.000	0.003	Chemokine	YES	YES	YES
PPBP	CXCR2	0.617	0.000	0.003	Chemokine	YES	YES	YES
CXCL8	CXCR1	0.607	0.000	0.003	Chemokine	YES	YES	YES
IL1RN	IL1R2	~0.593	~0.000	<0.05	Cytokine/JAK	NO	NO	YES
PDGFD	PDGFRB	0.583	0.000	0.002	PDGF	NO	NO	YES
IL12A	IL12RB2	0.547	0.092	0.010	IL-12/Th1	NO	NO	YES
CXCL5	CXCR2	0.584	0.000	0.003	Chemokine	YES	YES	YES
NCAM1	FGFR1	0.569	0.000	0.003	Cell adhesion	YES	YES	YES
AKR1C3	PTGDR	~0.430	~0.000	<0.05	Prostaglandin	NO	NO	NO
IFNL1	IL10RB	~0.420	~0.000	<0.05	IFN-lambda	NO	NO	NO

Top channel IFNL1->IFNLR1 (IFN-lambda) is invisible to LIANA, CellChat, AND NicheNet. IL1RN->IL1R2 is NEUROSCOPE-only in both Kang 2018 and Lupus — cross-disease IL-1 axis. 19/33 (58%) not in LIANA. Spectral gap 0.891 is the highest across all 4 benchmarks. ARI = 0.008 — communication states are essentially independent of transcriptomic clusters.

Smillie IBD Colon (Ulcerative colitis, n=18, 17 FDR-sig / 34 tested)

Ligand	Receptor	Capacity	CI lo	FDR	Pathway	LIANA	CellChat	NicheNet
LCK	CD8A	0.631	0.000	0.0008	T-cell	NO	NO	NO
IL1B	IL1R2	0.624	0.000	0.031	Cytokine/JAK	NO	NO	YES
VEGFA	NRP1	0.583	0.000	0.0005	EGF/ERBB	YES	YES	YES
IFNG	IFNGR1	0.445	0.000	0.0001	IFN-gamma	NO	NO	YES
THBS1	CD36	0.437	0.000	0.0015	Other	YES	YES	YES
KLRB1	CLEC2D	0.382	0.000	<0.0001	Other (NK)	NO	NO	NO
ICAM1	ITGB2	0.355	0.000	<0.0001	Cell adhesion	NO	NO	YES
ANXA1	FPR3	0.334	0.000	0.026	Resolution	NO	NO	NO
CLU	TYROBP	0.340	0.000	0.020	Complement	NO	NO	YES
APLP2	PIGR	0.330	0.000	0.018	Transcytosis	NO	NO	NO
PLAU	PLAUR	0.330	0.000	0.022	Fibrinolysis	YES	YES	YES
CXCL9	CXCR3	0.205	0.000	0.040	Chemokine	YES	YES	YES

12/17 (71%) NEUROSCOPE-significant pairs not in LIANA rank_aggregate — highest across all datasets. LCK->CD8A (top channel) is invisible to LIANA, CellChat, and NicheNet. 7/34 (20.6%) causally validated by 2SLS — highest causal validation rate (larger per-donor effect size). ARI = 0.031 — 100% of cells classified as divergent between comm and transcriptome.

R4 — Cross-Disease Findings

Pairs that are NEUROSCOPE-significant in multiple independent datasets, and invisible to all three comparison methods in each case.

L-R Pair	Datasets	Capacity range	LIANA	CellChat	NicheNet	Biological significance
IL1RN -> IL1R2	Kang 2018 + Lupus	0.593–0.913	NO	NO	NO	IL-1 decoy receptor; anakinra target (IFN-beta PBMC + SLE PBMC)
IL1B -> IL1R2	Lupus + Smillie IBD	0.593–0.624	NO	NO	YES	IL-1 beta signalling; key inflammatory axis (SLE PBMC + UC colon)
KLRB1 -> CLEC2D	Lupus + Smillie IBD	0.382–0.437	NO	NO	NO	NK lectin-type checkpoint receptor (SLE PBMC + UC colon)
CLU -> TYROBP	Kang 2018 + Smillie IBD	0.340–0.755	NO	NO	NO	Complement regulator + DAP12 signalling (IFN-beta PBMC + UC colon)

Yellow = invisible to that method. IL1RN->IL1R2 is the most striking finding: this pair is NEUROSCOPE-significant in both IFN-beta PBMC (Kang) and SLE PBMC (Lupus), across two independent diseases and cohorts, and is invisible to LIANA, CellChat, AND NicheNet in both. The mechanistic reason is the CellChat structural zero-floor: IL1RN is expressed in fewer than 50% of cells in most clusters, so trimean(IL1RN) = 0 in all cluster-pairs, and the pair is never tested.

R5 — Mechanism: Why KSG Detects More Than CellChat

The CellChat zero-floor is a mathematical consequence of the trimean formula, not a threshold choice. It produces a structural blind spot for any gene expressed in fewer than 50% of cells — which is extremely common in scRNA-seq data.

```
CellChat trimean: tm(x) = 0.25*Q1 + 0.5*Q2 + 0.25*Q3 (Jin et al. 2021, Nat Commun, eq. 1)
tm(x) = 0 whenever Q1 = Q2 = 0
Q1 = Q2 = 0 whenever fewer than 50% of cells express the gene (count > 0)
=> sparsity > 50% causes a HARD ZERO regardless of inter-donor variation
=> this is structural (formula property), not a parameter choice
=> it cannot be fixed by adjusting thresholds
NEUROSCOPE KSG: X_i = mean expression in cluster c, donor i (one value per donor)
MI = f(X_1,...,X_n, Y_1,...,Y_n) via k-NN density estimation
=> X_i > 0 whenever any cell in (cluster c, donor i) expresses the gene
=> inter-donor variation is preserved even at 10% per-cell sparsity
=> no expression threshold is applied before or after KSG estimation
Example (IL1RN in Kang 2018):
Per-cell sparsity: ~70% (most cells silent)
CellChat trimean: 0.000 in ALL cluster pairs -> pair never tested (zero-floor)
NEUROSCOPE: per-donor mean in CD14+ Mono: 0.08-0.43 (varies by donor)
KSG MI = 0.913 (FDR = 0.028) -> significant
This donor-level variation is statistically real and biologically valid
```

R6 — Communication Identity: Orthogonal to Transcriptome

Communication state identity (CSI) measures how well a cell's communication program matches its transcriptomic cluster assignment. ARI (Adjusted Rand Index) between NMF-derived communication states and transcriptomic cluster

labels. ARI = 1 means perfectly aligned; ARI = 0 means independent (random overlap).

Dataset	ARI (comm vs transcr.)	Classification	Spectral gap (communities)	Interpretation
Kang 2018	n/a	—	0.733	Comm identity not computed at n=16
COVID Atlas	0.082	DECOUPLED	0.812	100% cells classified as divergent
Lupus	0.008	DECOUPLED	0.891	Essentially zero — strongest decoupling
Smillie IBD	0.031	DECOUPLED	0.673	100% cells classified as divergent

Replicated across COVID Atlas, Lupus, and Smillie IBD: in all three datasets, a cell's communication behavior CANNOT be predicted from its transcriptomic cluster label. This means standard cell-type annotation is insufficient to characterize communication — NEUROSCOPE's information-theoretic channel representation captures a distinct and complementary biological dimension. The spectral gap > 0.67 in all datasets confirms that the communication network consistently separates into exactly 2 communities.

R7 — 2SLS Causal Validation

Two-stage least squares (2SLS) with donor_id as a natural instrumental variable (analogous to Mendelian Randomization). First stage: ligand ~ donor_id. Second stage: receptor ~ fitted_ligand. Weak instrument filter: first-stage F > 10. Second-stage p < 0.05. This tests whether ligand variation CAUSES receptor variation across donors, beyond mere statistical correlation.

Dataset	Donors (n)	Pairs tested	Causally validated	Rate	Note
Kang 2018	16	9	0	0%	Underpowered: n=16 insufficient for first-stage F>10
COVID Atlas	124	127	1	0.8%	n=124 barely sufficient; wide CIs expected
Lupus	149	40	1	2.5%	rank-deficient warnings expected at n=149 with 11 clusters
Smillie IBD	18	34	7	20.6%	Highest rate — larger per-donor effect size in UC

2SLS power scales with n_donors. Kang (n=16) is underpowered by design — this benchmark was chosen because it has ground truth (IFN-beta condition is the known intervention), not because it has causal power. Smillie IBD (n=18) achieves 20.6% causal validation despite small n, suggesting large per-donor effect sizes in UC. The 2SLS instrument (donor_id) is exogenous by design: donors are independent individuals whose genetic and environmental backgrounds causally affect ligand expression independently of which cell types they happen to have.

Methods

Channel capacity estimation

KSG k-NN mutual information estimator (Kraskov, Stögbauer & Grassberger 2004, Phys Rev E 69:066138). I(X;Y) = ψ(k) + ψ(n) − ■ψ(n■+1) + ψ(n■+1)■. Capacity = I(ligand ; response) / H(ligand) ∈ [0, 1]. Input: per-donor cluster means (pseudobulk). Adaptive k (default k=3, floored at ■n/5■). Spearman copula fallback when n < 10.

Permutation test

199 random shuffles of the receptor vector Y. Empirical p = fraction of null MI ≥ observed MI. BH-FDR correction across all tested L-R pairs.

Bootstrap confidence intervals

500 resamples (with replacement) of the n paired donor observations. 95% CI = [2.5th, 97.5th] percentile of bootstrap capacity distribution. Adaptive k applied consistently within each resample.

Normalization

Library-size median-ratio (scran-style): size factor = cell library size / median. Followed by log1p transform.

HVG selection

Degree-2 polynomial regression log(variance) ~ log(mean). Top 3000 genes by residual variance (Pearson residuals).

Differential channel testing

Two-sample Wilcoxon rank-sum on per-cluster-pair strengths (strength test) and capacities (capacity test). BH-FDR correction.

NMF programs

Non-negative matrix factorization of the [cluster-pair × L-R] communication tensor. Rank selected by elbow (<10% reconstruction improvement).

Causal validation (IV/2SLS)

Two-stage least squares with donor_id as natural instrumental variable (analogous to Mendelian Randomization in GWAS). Weak instrument filter: first-stage F > 10. Second-stage p < 0.05 required.

CellChat comparison

CellChat v1 (Jin et al. 2021 Nat Commun). Trimean statistic: 0.25·Q1 + 0.5·Q2 + 0.25·Q3. 1000 permutations. FDR < 0.05 threshold. Zero-floor: trimean = 0 when Q1 = Q2 = 0 (>50% sparsity).

CellPhoneDB comparison

CellPhoneDB v3 (Efremova et al. 2020 Nat Protoc). Mean(L in cluster A) × mean(R in cluster B), min 10% expression, 1000 permutations. FDR < 0.05.

NicheNet comparison

NicheNet v2 (Browaeys et al. 2020 Nat Methods; nichenetr R package). Ligand activity = AUPR over receiver DE gene set enrichment via curated TF-target, PPI, and signaling networks from Zenodo (record 7074291). Run: Rscript compare_nichenet.R. Top-20 NicheNet ligands compared to NEUROSCOPE FDR-significant pairs.

LIANA multi-method comparison

LIANA (Dimitrov et al. 2022 Nat Cell Biol; liana-py ≥1.0). Wraps CellPhoneDB, CellChat, NicheNet, NATMI, and Connectome. rank_aggregate: consensus Borda-count rank across all methods. expr_prop = 0.1 (min 10% expressing cells). Run: python compare_liana.py. Install: pip install liana anndata scanpy.

Software

MATLAB R2021b+. Statistics and Machine Learning Toolbox. Python 3.x for data preparation (anndata, scanpy, scipy, numpy, pandas, liana). R 4.x for NicheNet comparison (nichenetr, tidyverse, Seurat).